

CUSTOMER SUPPORT CHATBOT USING AMAZON LEX

ABSTRACT

In recent years, chatbots have become an important part of customer support systems to provide quick and automated responses to users. This project focuses on the development of a Customer Support Chatbot using Amazon Lex V2 and AWS Lambda. The chatbot is designed to handle common customer support requests such as greeting users, resetting passwords, and creating support tickets.

Amazon Lex is used to understand user input, identify intents, and manage conversation flow using sample utterances and slots. AWS Lambda is integrated with Amazon Lex to handle backend processing and intent fulfillment in a serverless manner. The chatbot collects required information such as email address or issue description and provides appropriate responses to the user.

The system follows a scalable and cost-effective serverless architecture, reducing the need for manual customer support intervention. Although advanced features like email notifications and database storage are not implemented, the project demonstrates a clear understanding of conversational AI, cloud services, and real-time interaction using AWS. This chatbot can be further enhanced by integrating services such as Amazon SES and DynamoDB for real-world deployment.

1. INTRODUCTION

In today's digital world, customer support plays a very important role in improving user experience. Many organizations use chatbots to handle common customer queries automatically without human intervention. This project focuses on building a **Customer Support Chatbot** using **Amazon Lex V2** and **AWS Lambda** to handle basic customer support functionalities.

The chatbot can greet users, help them reset their password, and allow them to raise a support ticket by collecting issue details. This system reduces manual support effort and provides instant responses to users.

2. OBJECTIVE OF THE PROJECT

The main objectives of this project are:

- To build a cloud-based chatbot using AWS services
 - To understand how Amazon Lex intents, slots, and fulfillment work
 - To integrate AWS Lambda with Amazon Lex
 - To handle common customer support use cases automatically
 - To gain hands-on experience with real-time cloud services
-

3. AWS SERVICES USED

3.1 Amazon Lex V2

Amazon Lex is a service for building conversational interfaces using voice and text. It is used to:

- Understand user input
- Identify user intent
- Manage conversation flow using intents and slots

3.2 AWS Lambda

AWS Lambda is a serverless compute service. It is used to:

- Process backend logic
- Handle fulfillment of intents
- Generate dynamic responses

3.3 IAM (Identity and Access Management)

IAM is used to:

- Grant permissions for Lex to invoke Lambda
 - Secure interaction between AWS services
-

4. SYSTEM ARCHITECTURE

4.1 Architecture Description

The architecture of the chatbot follows a simple serverless design:

1. User interacts with the chatbot through text
2. Amazon Lex analyzes the input
3. Lex identifies the intent
4. Lex triggers AWS Lambda (if fulfillment is enabled)
5. Lambda processes the request and sends response
6. Amazon Lex sends final response to the user

4.2 Architecture Flow

User → Amazon Lex → AWS Lambda → Amazon Lex → User

5. WORKING FLOW (STEP-BY-STEP)

Step 1: User Input

The user types a message such as:

- "reset my password"
- "report an issue"

Step 2: Intent Recognition

Amazon Lex matches the user input with sample utterances and identifies the correct intent.

Step 3: Slot Collection

If required, Lex asks the user for additional information such as:

- Registered email address
- Issue description

Step 4: Fulfillment

Once all required data is collected:

- Lex triggers AWS Lambda
- Lambda processes the request

Step 5: Response

The chatbot sends a confirmation message to the user.

6. INTENTS IMPLEMENTED

6.1 GreetingIntent

Purpose: Greet the user

Sample Utterances:

- hi
- hello
- hey

Response:

"Hello 🙌 How can I help you today?"

6.2 PasswordResetIntent

Purpose: Handle password reset requests

Sample Utterances:

- reset my password
- I forgot my password
- password reset

Slots Used:

- Email (Amazon.FreeFormInput)

Fulfillment:

- AWS Lambda

Final Response:

"Your password reset request is received. A reset link will be sent to your email."

6.3 TicketCreationIntent

Purpose: Raise a support ticket

Sample Utterances:

- report an issue
- need support
- app not working

Slots Used:

- IssueDescription (Amazon.FreeFormInput)

Fulfillment:

- AWS Lambda

Final Response:

"Your issue has been recorded. Our support team will contact you shortly."

7. USE CASES

Use Case 1: Password Reset

- User requests password reset
- Bot asks for email
- Bot confirms request

Use Case 2: Ticket Creation

- User reports an issue
- Bot collects issue description
- Bot confirms ticket creation

Use Case 3: Greeting

- User greets the bot
 - Bot responds with greeting
-

8. SAMPLE OUTPUT

User: reset my password

Bot: Please enter your registered email address

User: user@gmail.com

Bot: Your password reset request is received. A reset link will be sent to your email.

9. ADVANTAGES OF THE SYSTEM

- Reduces manual customer support workload
 - Instant response to users
 - Serverless and scalable architecture
 - Easy to enhance with new features
-

10. LIMITATIONS

- Email sending not implemented (SES not integrated)
 - No database storage
 - Limited to basic support features
-

11. FUTURE ENHANCEMENTS

- Integrate Amazon SES for real email notifications
 - Store tickets using Amazon DynamoDB
 - Add multi-language support
 - Deploy bot on web or WhatsApp
 - Add authentication and user verification
-

12. CONCLUSION

This project successfully demonstrates the creation of a Customer Support Chatbot using Amazon Lex and AWS Lambda. It provides hands-on experience with cloud services, serverless architecture, and conversational AI. The chatbot efficiently handles basic customer support scenarios and can be extended further for real-world applications.

13. REFERENCES

- [AWS Documentation – Amazon Lex](#)
- [AWS Documentation – AWS Lambda](#)
- [AWS IAM Documentation](#)